

Lecture 6: Induction

September 10, 2026

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1 Recall: The Fifth Peano Axiom

In Lecture 2, we built the natural numbers from 0 and a successor operation S . The first four Peano axioms gave us a chain starting at 0, but still allowed extra, disconnected chains. The fifth axiom, **induction**, ruled out those extra objects. Intuitively, every natural number lies on the chain

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \dots$$

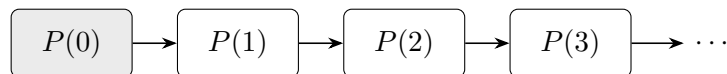
As before, $\mathbb{N} = \{0, 1, 2, \dots\}$.

The fifth axiom says that if a property holds at 0, and whenever it holds at a natural number it also holds at its successor, then it holds at every natural number. Today we use this axiom as a method of proof.

Let $P(n)$ be the statement we want to prove about n . There are two things to establish:

1. **Base case:** prove $P(0)$.
2. **Inductive step:** for an arbitrary $i \in \mathbb{N}$, assume $P(i)$ and prove $P(i + 1)$.

The temporary assumption $P(i)$ is called the **induction hypothesis**. Once both parts are proved, induction gives $P(n)$ for every $n \in \mathbb{N}$.



Base case: prove $P(0)$	Inductive step: prove $P(i) \Rightarrow P(i + 1)$ for every $i \geq 0$
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The base case starts the argument. The inductive step lets us move from each item on the chain to the next. Thus $P(0)$ gives $P(1)$, which gives $P(2)$, and so on. We do not need to write infinitely many separate proofs: we prove one implication that works at *every* link.

In the inductive step, we do not assume the theorem for every n , or assume the conclusion $P(i + 1)$. We prove a conditional statement: *if* $P(i)$ holds, *then* $P(i + 1)$ holds. The base case gives the first unconditional fact from which the chain can start.

Both parts are necessary. For example, let $P(n)$ be the false statement $n = n + 1$. Assuming $i = i + 1$ and adding 1 gives $i + 1 = i + 2$, so the inductive step works. But the base case $0 = 1$ fails, and induction therefore gives no conclusion. Likewise, checking a few starting values without proving the inductive step does not establish a claim for all n .

2 Examples

Example 1: The Sum of the First n Integers

Claim 1. For every integer $n \geq 0$,

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

The notation $\sum_{k=1}^n k$ means $1 + 2 + \cdots + n$ when $n \geq 1$. When $n = 0$, there are no terms; we define this **empty sum** to be 0. This convention lets us start the proof at 0.

Proof by induction. **Base case** ($n = 0$). The sum is 0, and $0(0+1)/2 = 0$, so the formula holds.

Inductive step. Let $i \geq 0$ be arbitrary and assume

$$\sum_{k=1}^i k = \frac{i(i+1)}{2}.$$

We must prove the formula with $i+1$ in place of i . Separate off the last term and use the induction hypothesis:

$$\begin{aligned} \sum_{k=1}^{i+1} k &= \left(\sum_{k=1}^i k \right) + (i+1) \\ &= \frac{i(i+1)}{2} + (i+1) && \text{(induction hypothesis)} \\ &= \frac{i(i+1) + 2(i+1)}{2} \\ &= \frac{(i+1)(i+2)}{2}. \end{aligned}$$

This is exactly the desired formula for $i+1$. By induction, the identity holds for every integer $n \geq 0$. $\square$

Gauss's Pairing Argument.

There is another way to prove the same formula. For $n \geq 1$, let $T = 1 + 2 + \cdots + n$. Write the sum once forwards and once backwards:

$$\begin{array}{rccccccc} T & = & 1 & + & 2 & + & \cdots + & n \\ T & = & n & + & (n-1) & + & \cdots + & 1. \end{array}$$

Adding corresponding terms gives n copies of $n+1$. Therefore

$$2T = n(n+1), \quad T = \frac{n(n+1)}{2}.$$

For example, $1 + 2 + \cdots + 100 = 100 \cdot 101/2 = 5050$. The case $n = 0$ again gives $0 = 0$.

There can be several good proofs of the same statement. Induction verifies the formula one step at a time; pairing explains it by rearranging the sum. Choose the method that makes the argument clearest.

Example 2: The Sum of the First n Squares

Claim 2. For every integer $n \geq 0$,

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof by induction. **Base case** ($n = 0$). Both sides are 0.

Inductive step. Let $i \geq 0$ be arbitrary and assume

$$\sum_{k=1}^i k^2 = \frac{i(i+1)(2i+1)}{6}.$$

The next sum has one additional term, $(i+1)^2$. Hence

$$\begin{aligned} \sum_{k=1}^{i+1} k^2 &= \left(\sum_{k=1}^i k^2 \right) + (i+1)^2 \\ &= \frac{i(i+1)(2i+1)}{6} + (i+1)^2 && \text{(induction hypothesis)} \\ &= \frac{(i+1)(2i^2 + 7i + 6)}{6} \\ &= \frac{(i+1)(i+2)(2i+3)}{6}. \end{aligned}$$

This is the formula for $i+1$, so the claim follows by induction. □

Before doing the algebra, write down the target formula for $i+1$. Here the new term is $(i+1)^2$, and the target factors are $(i+1)$, $(i+2)$, and $(2i+3)$.

Example 3: A Divisibility Claim

Claim 3. For every integer $n \geq 0$, the number $6^n - 1$ is divisible by 5.

We write $5 \mid (6^n - 1)$ to mean that $6^n - 1 = 5q$ for some integer q .

Proof by induction. **Base case** ($n = 0$). We have $6^0 - 1 = 0 = 5 \cdot 0$, so the claim holds. Zero is divisible by 5.

Inductive step. Let $i \geq 0$ be arbitrary and assume $5 \mid (6^i - 1)$. Then $6^i - 1 = 5q$ for some integer q . Rewrite the next expression so that the old one appears:

$$\begin{aligned} 6^{i+1} - 1 &= 6 \cdot 6^i - 1 \\ &= 6(6^i - 1) + 5 \\ &= 6(5q) + 5 && \text{(induction hypothesis)} \\ &= 5(6q + 1). \end{aligned}$$

Since $6q + 1$ is an integer, $6^{i+1} - 1$ is divisible by 5. Induction proves the claim for every integer $n \geq 0$. $\square$

The useful step was rewriting $6^{i+1} - 1$ in terms of $6^i - 1$. An induction hypothesis only helps if we find a place to use it.

Example 4: Postage Using 5-Cent and 8-Cent Stamps

Suppose we need to pay an integer number of cents in postage and have an unlimited supply of 5-cent stamps and 8-cent stamps. We want to pay the exact amount, using only these denominations.

Claim 4. *Every integer amount $n > 28$ cents can be written as*

$$n = 5a + 8b \quad \text{for some integers } a, b \geq 0.$$

The numbers a and b count the stamps of each type, so neither can be negative. Let $P(n)$ mean that n cents can be made.

Proof by exchanging stamps. We will prove the slightly stronger statement that every integer amount of at least 28 cents can be made. The key observations are

$$3 \cdot 8 + 1 = 5 \cdot 5 \quad \text{and} \quad 3 \cdot 5 + 1 = 2 \cdot 8.$$

Thus replacing three stamps of either denomination with an appropriate number of the other denomination increases the total by exactly one cent.

Base case ($n = 28$). We have $28 = 4 \cdot 5 + 8$.

Inductive step. Let $i \geq 28$ and assume that $i = 5a + 8b$ for some nonnegative integers a, b . If there were at most two stamps of each denomination, then

$$i \leq 2 \cdot 5 + 2 \cdot 8 = 26 < 28,$$

a contradiction. Therefore $a \geq 3$ or $b \geq 3$.

- If $a \geq 3$, replace three 5-cent stamps with two 8-cent stamps:

$$i + 1 = 5(a - 3) + 8(b + 2).$$

- Otherwise, $b \geq 3$. Replace three 8-cent stamps with five 5-cent stamps:

$$i + 1 = 5(a + 5) + 8(b - 3).$$

In either case, the new stamp counts are nonnegative and the total is $i + 1$ cents. Induction proves $P(n)$ for every integer $n \geq 28$, and hence for every integer $n > 28$ as claimed. $\square$

Example 5: Counting Subsets

In Lecture 5, we represented subsets by choosing which elements to include. Induction gives another proof of the resulting count. Our induction statement will cover *every* set of a given size.

Claim 5. *Every finite set with n elements has exactly 2^n subsets.*

Proof by induction on n . **Base case** ($n = 0$). The empty set has exactly one subset, namely itself. Thus $|\mathcal{P}(\emptyset)| = 1 = 2^0$.

Inductive step. Assume that every set with i elements has 2^i subsets. Let A be any set with $i + 1$ elements. Choose an element $a \in A$ and put $B = A \setminus \{a\}$. Then B has i elements, so the induction hypothesis says it has 2^i subsets.

Partition the subsets of A into two groups:

1. Those that **do not contain** a . These are exactly the subsets of B , so there are 2^i of them.
2. Those that **contain** a . Each is obtained uniquely by taking a subset of B and adding a , so there are also 2^i of them.

The groups are disjoint and include every subset of A . Therefore

$$|\mathcal{P}(A)| = 2^i + 2^i = 2^{i+1}.$$

This proves the claim for every set of size $i + 1$, completing the induction. $\square$